

Review of Chapter 7.

§7.1.

1. Laplace Transform of a function $f(t)$ ($t \geq 0$)

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt \Leftrightarrow f(t) = \mathcal{L}^{-1}\{F(s)\}.$$

Linearity

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}.$$

2.

$f(t)$	$F(s)$
1	$\frac{1}{s} \quad (s > 0)$
t	$\frac{1}{s^2} \quad (s > 0)$
t^n	$\frac{n!}{s^{n+1}} \quad (s > 0)$
$t^a \quad (a > -1)$	$\frac{\Gamma(a+1)}{s^{a+1}} \quad (s > 0)$
e^{at}	$\frac{1}{s-a} \quad (s > a)$
$\cos kt$	$\frac{s}{s^2+k^2} \quad (s > 0)$
$\sin kt$	$\frac{k}{s^2+k^2} \quad (s > 0)$
$\cosh kt$	$\frac{s}{s^2-k^2} \quad (s > k)$
$\sinh kt$	$\frac{k}{s^2-k^2} \quad (s > k)$
$u(t-a)$	$\frac{e^{-as}}{s} \quad (s > 0)$

unit step function

$$u(t-a) = \begin{cases} 0 & t < a \\ 1 & t \geq a. \end{cases}$$

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt.$$

$$\Gamma(n+1) = n!$$

§7.2.

1. $ax'' + bx' + cx = f$

$$\Rightarrow \mathcal{L}\{ax'' + bx' + cx\} = \mathcal{L}\{f\} \Rightarrow a\mathcal{L}\{x''\} + b\mathcal{L}\{x'\} + c\mathcal{L}\{x\} = \mathcal{L}\{f\}$$

$\mathcal{L}\{x(t)\} = \underline{X}(s)$, then

Transform of Derivatives	}	$\mathcal{L}\{x'(t)\} = s\mathcal{L}\{x(t)\} - x(0) = s\underline{X}(s) - x(0)$
		$\mathcal{L}\{x''(t)\} = s^2\mathcal{L}\{x(t)\} - sx(0) - x'(0) = s^2\underline{X}(s) - sx(0) - x'(0)$
		$\vdots$
		$\mathcal{L}\{x^{(n)}(t)\} = s^n\mathcal{L}\{x(t)\} - s^{n-1}x(0) - s^{n-2}x'(0) - \dots - sx^{(n-2)}(0) - x^{(n-1)}(0)$ $= s^n\underline{X}(s) - s^{n-1}x(0) - s^{n-2}x'(0) - \dots - sx^{(n-2)}(0) - x^{(n-1)}(0)$

2. Solution process of the initial value problem:

$$ax'' + bx' + cx = f; \quad x(0) = x_0, \quad x'(0) = x_0'$$

Step 1: Let $\mathcal{L}\{x(t)\} = \underline{X}(s)$, compute

$$\mathcal{L}\{x'(t)\} = s\underline{X}(s) - x_0$$

$$\mathcal{L}\{x''(t)\} = s^2\underline{X}(s) - sx_0 - x_0'$$

Step 2: set up $\mathcal{L}\{ax'' + bx' + cx\} = \mathcal{L}\{f\}$, i.e.

$$a\mathcal{L}\{x''\} + b\mathcal{L}\{x'\} + c\mathcal{L}\{x\} = \mathcal{L}\{f\}$$

Solve the algebraic equation to get $\underline{X}(s)$.

Step 3: Find $x(t) = \mathcal{L}^{-1}\{\underline{X}(s)\}$ by inverse Laplace transform.

3. Transform of Integrals

$F(s) = \mathcal{L}\{f(t)\}$, then

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s} \mathcal{L}\{f(t)\} = \frac{F(s)}{s}$$

or equivalently,

$$\mathcal{L}^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(\tau) d\tau$$

$$4. \mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}, \quad \mathcal{L}\{t \cos kt\} = \frac{s^2 - k^2}{(s^2 + k^2)^2}$$

§7.3.

1. Rational function

$$R(s) = \frac{P(s)}{Q(s)}$$

Partial fraction decomposition:

a. Linear factor

$$\frac{A_1}{s-a} + \frac{A_2}{(s-a)^2} + \dots + \frac{A_n}{(s-a)^n}$$

b. Quadratic factor

$$\frac{A_1 s + B_1}{(s-a)^2 + b^2} + \frac{A_2 s + B_2}{[(s-a)^2 + b^2]^2} + \dots + \frac{A_n s + B_n}{[(s-a)^2 + b^2]^n}$$

2. Translation on the s-Axis

let $F(s) = \mathcal{L}\{f(t)\}$, then

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

or equivalently,

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$$

$f(t)$	$F(s)$
$e^{at}t^n$	$\frac{n!}{(s-a)^{n+1}}$
$e^{at}\cos kt$	$\frac{s-a}{(s-a)^2 + k^2}$
$e^{at}\sin kt$	$\frac{k}{(s-a)^2 + k^2}$

$$3. \mathcal{L}^{-1}\left\{\frac{s}{(s^2+k^2)^2}\right\} = \frac{1}{2k} t \sin kt$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+k^2)^2}\right\} = \frac{1}{2k^3}(\sin kt - kt \cos kt)$$

§7.4.

1. Convolution of two functions f and g .

$$(f * g)(t) = \int_0^t f(\tau) g(t-\tau) d\tau.$$

Remark: $f * g = g * f$.

2. The Convolution Property, let $F(s) = \mathcal{L}\{f(t)\}$, $G(s) = \mathcal{L}\{g(t)\}$

$$\mathcal{L}\{f(t) * g(t)\} = \mathcal{L}\{f(t)\} \cdot \mathcal{L}\{g(t)\} = F(s)G(s)$$

or equivalently,

$$\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$$

3. Differentiation of Transforms

let $F(s) = \mathcal{L}\{f(t)\}$, then

$$\mathcal{L}\{-tf(t)\} = F'(s)$$

or equivalently,

$$f(t) = -\frac{1}{t} \mathcal{L}^{-1}\{F'(s)\}.$$

Remark:

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$$

4. Integration of Transforms.

let $F(s) = \mathcal{L}\{f(t)\}$, then

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^{+\infty} F(\sigma) d\sigma$$

or equivalently,

$$f(t) = t \mathcal{L}^{-1}\left\{\int_s^{+\infty} F(\sigma) d\sigma\right\}.$$